

# Arithmetic and Order

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- **Exercise 1 (Additive Cancellation)**

Let  $a, b, c \in \mathbb{Z}$ . Prove that  $a + b = a + c \implies b = c$ .

- **Proof 1.1**

Suppose  $a, b, c \in \mathbb{Z}, a + b = a + c$ . Consider the additive inverse of  $-a$  of  $a$ .

$$\begin{aligned} -a + (a + b) &= -a + (a + c) \\ (-a + a) + b &= (-a + a) + c \\ 0 + b &= 0 + c \\ b &= c. \end{aligned}$$

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- **Exercise 2 (If adding something changes nothing)**

Suppose  $a, b \in \mathbb{Z}$  and  $a + b = a$ . Prove that  $b = 0$ .

- **Proof 2.1**

Suppose  $a, b \in \mathbb{Z}$  and  $a + b = a$ . By properties,

$$\begin{aligned} a + 0 &= a \\ a + b &= a + 0 \\ b &= 0. \end{aligned}$$

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- **Exercise 3**

Suppose  $a, b \in \mathbb{Z}$  and  $a + b = 0$ . Prove that  $b = -a$ .

- **Proof 3.1**

Suppose  $a, b \in \mathbb{Z}$  and  $a + b = 0$ . By properties,

$$\begin{aligned} a + (-a) &= 0 \\ a + b &= a + (-a) \\ b &= -a. \end{aligned}$$

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- **Exercise 4**

Suppose  $a, b, x, y \in \mathbb{Z}, a + x = b \wedge a + y = b$ . Prove  $x = y$ .

- **Proof 4.1**

Suppose  $a, b, x, y \in \mathbb{Z}, a + x = b, a + y = b$ . Then  $a + x = a + y$ . By cancellation law  $x = y$ .

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- **Exercise 5**

Suppose  $e \in \mathbb{Z}$  has the property that  $\forall a \in \mathbb{Z}, a + e = a$ . Prove that  $e = 0$ .

- **Proof 5.1**

Suppose  $e, a \in \mathbb{Z}, a + e = a = a + 0 \implies e = 0$ .

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- **Exercise 6**

Suppose  $a, b, c \in \mathbb{Z}, a + b = 0 \wedge a + c = 0$ . Prove that  $b = c$ .

- **Proof 6.1**

Suppose  $a, b, c \in \mathbb{Z}, a + b = 0 \wedge a + c = 0$ . Then,  $a + c = a + b$ . By cancellation law,  $c = b$ .

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- **Exercise 7**

Prove that  $-0 = 0$ .

◦ **Proof 7.1**

$0 + (-0) = 0 = 0 + 0$ . By cancellation law,  $-0 = 0$ .

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• **Exercise 8 (Negating an inequality)**

Suppose  $a, b \in \mathbb{Z}, a < b \implies -b < -a$ .

◦ **Proof 8.1**

Suppose  $a, b \in \mathbb{Z}, a < b$ . By definition,  $a < b \iff a \leq b \wedge a \neq b$ . Then

$$\begin{aligned} a + (-a - b) &\leq b + (-a - b) \\ (a + (-a)) + (-b) &\leq (b + (-b)) + (-a) \\ 0 + (-b) &\leq 0 + (-a) \\ -b &\leq -a \end{aligned}$$

As  $-a \neq -b \implies -b < -a$ .

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• **Exercise 9**

Suppose  $a, b \in \mathbb{Z}, c < 0, a < b$  Prove that  $ac > bc$ .

◦ **Proof 9.1**

Suppose  $a, b \in \mathbb{Z}, c < 0, a < b$ . By definition,  $a < b \iff a \leq b \wedge a \neq b$ . Then

$$\begin{aligned} -c > 0 &\implies (-c)a \leq (-c)b \\ &\implies -ca \leq -cb \\ &\implies -ca < -cb \quad (\text{as } a \neq b \implies ca \neq cb \implies -ca \neq -cb) \\ &\implies ca > cb \end{aligned}$$